

Tech Academy Solution

DATA ANALYTICS & BUSINESS INTELLIGENCE CAPSTONE PROJECT GUIDELINES

Introduction

Dear Students,

As part of your training in **Data Analytics and Business Intelligence**, you will be working in groups of **2–3 students** to complete a comprehensive Capstone Project using real-world datasets from different industries.

This Capstone Project is designed to simulate real-world business environments where analysts are expected to:

- Clean and prepare raw data
- Analyze business performance
- Generate actionable insights
- Build interactive dashboards
- Present business recommendations to stakeholders

The project will test both your **Data Analytics** and **Business Intelligence** skills using tools such as:

- Excel
- SQL
- Python
- Power BI

Each group is expected to select ONE dataset/project topic from the list provided below and complete the full end-to-end analytics workflow.

CAPSTONE PROJECT OBJECTIVES

The objective of this project is to help students:

- Apply real-world analytical techniques
- Develop problem-solving skills
- Build professional dashboards
- Practice business storytelling
- Generate data-driven recommendations
- Work collaboratively as analysts

- Improve presentation and communication skills
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GROUP REQUIREMENTS

- Students must work in groups of **2–3**
 - Each group must select **ONLY ONE dataset/project**
 - Every student must contribute actively
 - Professionalism, teamwork, and originality are expected
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AVAILABLE PROJECT TOPICS & DATASETS

1. Healthcare / Patient Analytics

Analyze patient records, healthcare trends, treatment costs, and hospital performance.

Dataset:

[Healthcare Dataset](#)

2. Finance / Loan Risk Analytics

Analyze loan applications, borrower behavior, and default risk patterns.

Dataset:

[Loan Default Dataset](#)

3. Marketing Campaign Analytics

Analyze campaign effectiveness, conversion trends, customer engagement, and ROI.

Dataset:

[Marketing Campaign Dataset](#)

4. E-Commerce Customer Analytics

Analyze customer purchasing behavior, retention, and product performance.

Dataset:

[E-Commerce Customer Behavior Dataset](#)

5. Human Resources (HR) Analytics

Analyze employee attrition, workforce performance, and departmental trends.

Dataset:

[HR Analytics Dataset](#)

6. Food Delivery / Logistics Analytics

Analyze delivery performance, customer ratings, delays, and operational efficiency.

Dataset:

[Food Delivery Dataset](#)

7. Airline / Travel Analytics

Analyze passenger satisfaction, flight delays, cancellations, and operational trends.

Dataset:

[Airline Passenger Satisfaction Dataset](#)

8. Real Estate Analytics

Analyze housing prices, market trends, and regional property performance.

Dataset:

[House Prices Dataset](#)

CAPSTONE PROJECT REQUIREMENTS

Each group MUST complete all stages below.

PHASE 1 — BUSINESS UNDERSTANDING

Each group must:

- Understand the business problem
- Identify business goals
- Define analytical objectives
- Identify key business KPIs

Students must explain:

- What problem the business is facing
- What insights the company wants to discover
- Why the analysis matters

PHASE 2 — DATA CLEANING & PREPARATION

Students are expected to:

- Handle missing values
- Remove duplicates
- Correct inconsistencies
- Standardize formats
- Convert data types
- Create calculated columns where necessary
- Prepare data for analysis and visualization

Tools allowed:

- Excel
- SQL
- Python (Pandas)

Students must document:

- Cleaning steps performed
 - Assumptions made
 - Data quality issues identified
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PHASE 3 — DATA ANALYSIS

Students must perform meaningful business analysis such as:

- Trend Analysis
- Customer Analysis
- Profitability Analysis
- Regional Analysis
- Product Performance Analysis
- Correlation Analysis
- Operational Performance Analysis
- KPI Tracking

Students are expected to identify:

- Patterns
 - Trends
 - Opportunities
 - Risks
 - Business challenges
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PHASE 6 — BUSINESS INTELLIGENCE DASHBOARD (POWER BI)

Each group must create a professional and interactive Power BI dashboard.

The dashboard **MUST** include:

Interactive Visualizations

Examples:

- Bar Charts
- Line Charts
- Area Charts
- Maps
- Tables
- Slicers/Filters

Dashboard Requirements

- Professional design
- Proper alignment
- Clear storytelling
- Consistent formatting
- Business-focused visuals
- Interactive user experience

Students must ensure the dashboard:

- Answers business questions
- Supports decision-making
- Is easy to understand

PHASE 7 — BUSINESS INSIGHTS & RECOMMENDATIONS

This is one of the MOST IMPORTANT sections.

Each group must provide:

Key Findings

What did the analysis reveal?

Business Insights

Why do these findings matter?

Recommendations

What actions should the business take?

Recommendations must be:

- Practical
- Actionable
- Data-driven

Students must think like Business Analysts and Consultants.

PHASE 8 — PRESENTATION

Each group will present their project on the assigned presentation date.

The presentation should include:

1. Introduction to the dataset
2. Business problem
3. Cleaning process
4. Analysis performed
5. SQL queries used
6. Python EDA findings
7. Power BI dashboard walkthrough
8. Key insights
9. Recommendations
10. Challenges faced

All group members MUST participate.

FINAL DELIVERABLES

Each group must submit:

1. Cleaned Dataset
 2. SQL Queries/File
 3. Python Notebook (.ipynb)
 4. Power BI Dashboard (.pbix)
 5. PowerPoint Presentation
 6. Project Documentation/Report
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IMPORTANT NOTES

- Creativity and professionalism are highly encouraged
 - Focus on business storytelling, not only visuals
 - Dashboard quality and insight generation matter
 - Teams should think like professional analysts
 - Originality and teamwork are important
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FINAL MESSAGE

This Capstone Project is designed to simulate real-world Data Analytics and Business Intelligence workflows used in modern organizations.

Your goal is not only to analyze data but to transform data into meaningful business decisions and recommendations.

Approach this project with professionalism, creativity, analytical thinking, and teamwork.

I look forward to seeing your dashboards, insights, and presentations.

Best wishes to all groups.